

Robust Station Selection for Microtransit Under Demand Uncertainty

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1 Introduction

Microtransit is a form of transit that seeks to fill the gap between existing services such as private-hire vehicles and scheduled public transit. It is an on-demand, publicly accessible shared-ride service. Like private-hire, it is flexible and on-demand, but it also benefits from the economies of scale of scheduled public transportation by using mid-size capacity vehicles and allowing for greater vehicle utilization.

However, having a fully flexible microtransit system is overly complex and reduces its aggregation capabilities, often having to make significant detours just to satisfy one or two customers. Thus, selection of these *virtual bus stops* (VBS) becomes an important strategic decision. The optimal selection enables better passenger aggregation, reducing total vehicle distance travelled and improving vehicle utilization, while offering passengers decreased waiting times and fewer detours than scheduled public transit.

While the VBS selection is able to adapt between service periods within the day (morning, afternoon, evening, night), the selection for each service period should be robust against shifts in the demand patterns. Within a service period, demand can fluctuate from day to day and the system should still provide a relatively similar experience across each day. We thus address this problem with a two-stage robust model where origin-destination (OD) demand varies within a budgeted uncertainty set.

2 Model

We formulate VBS selection as a two-stage mixed-integer program. The first stage designates a fixed set of l stations shared across all service periods. The second stage activates k of these per period, and OD pair assignments are used to evaluate the demand-weighted walking and routing cost of each activation plan. We first present the nominal model assuming known demand, then characterise the demand uncertainty set, and finally derive the robust counterpart.

2.1 Nominal Model

Sets.

- J set of candidate stations,
- S set of service scenarios,
- Ω set of origin-destination pairs,
- A_{od} admissible pickup-dropoff station pairs for OD pair (o, d) ,

where

$$A_{od} = \left\{ (j, k) \in J \times J : d_{oj}^{\text{orig}} \leq W^{\text{max}}, d_{dk}^{\text{dest}} \leq W^{\text{max}} \right\}.$$

In the experimental instance, A_{od} is non-empty for all $(o, d) \in \Omega$.

Input parameters. The following parameters are assumed to be known:

$q_{ods} \geq 0$	demand for OD pair (o, d) in scenario s ,
$d_{oj}^{\text{orig}} \geq 0$	walking distance from origin o to station j ,
$d_{dk}^{\text{dest}} \geq 0$	walking distance from station k to destination d ,
$c_{jk} \geq 0$	vehicle travel cost from station j to station k ,
$\lambda \geq 0$	weight on vehicle travel cost relative to walking distance,
$W^{\text{max}} \geq 0$	maximum walking distance allowed for a passenger,
$l \in \mathbb{Z}_+$	number of stations to select,
$k \in \mathbb{Z}_+$	number of stations to activate per scenario.

Decision variables. The model has three layers of binary decisions:

$$\begin{aligned}
y_j &= \begin{cases} 1 & \text{station } j \text{ is selected} \\ 0 & \text{otherwise} \end{cases} \\
z_{js} &= \begin{cases} 1 & \text{station } j \text{ is activated and available for service in scenario } s \\ 0 & \text{otherwise} \end{cases} \\
x_{odjks} &= \begin{cases} 1 & \text{OD pair } (o, d) \text{ is assigned to pickup station } j \text{ and dropoff station } k \text{ in scenario } s \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

Objective. The objective minimizes total demand-weighted travel cost across all scenarios:

$$\min_{x, y, z} \sum_{s \in S} \sum_{(o, d) \in \Omega} \sum_{(j, k) \in A_{od}} q_{ods} a_{odjks} x_{odjks} \quad (1)$$

$$\text{where } a_{odjks} = d_{oj}^{\text{orig}} + d_{dk}^{\text{dest}} + \lambda c_{jk}$$

Constraints. The constraints enforce the two-stage structure:

$$\sum_{(j, k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s \quad (2)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s \quad (3)$$

$$z_{js} \leq y_j \quad \forall j, s \quad (4)$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \quad \forall s \quad (5)$$

Constraint (2) requires every OD pair to be assigned to exactly one admissible station pair per scenario. Constraints (3) ensure that assignments can only use active stations. Constraint (4) requires that a station must be selected before it can be activated. Constraint (5) fixes the number of selected stations to l and the number of active stations per scenario to k .

2.2 Robust Counterpart

The nominal model assumes demand q_{ods} is known exactly, but in practice it varies from day to day. To guard against this, we require the station selection to perform well under any demand realisation within a budget uncertainty set.

2.2.1 Demand Uncertainty

For each period s , demand lies in

$$\mathcal{U}_s = \left\{ q_s : \underline{q}_{ods} \leq q_{ods} \leq \bar{q}_{ods} \ \forall (o, d), \sum_{(o,d) \in \Omega} q_{ods} \leq \bar{Q}_s, q_{ods} \in \mathbb{Z}_+ \forall (o, d) \right\}. \quad (6)$$

The per-OD bounds $[\underline{q}_{ods}, \bar{q}_{ods}]$ reflect the observation that demand for a given OD pair fluctuates around a historical range from day to day. There is also a aggregate limit defined by $\bar{Q}_s$, which represents the idea that while demand can shift across OD pairs, the total number of trips within a time period is unlikely to increase beyond the population limit. Together, this uncertainty set is able to accurately represent worst case demand shifts over each scenario.

One concern is that since passenger demand is integral, this uncertainty set might be difficult to optimize over, but since the bounds $\underline{q}_{ods}, \bar{q}_{ods}, \bar{Q}_s$ should be integer values, the extreme points of the linear relaxation of the uncertainty set will also be integer. Thus, integrality can be dropped from the uncertainty set without changing the worst-case value.

2.2.2 Robust Objective and Recourse

Given fixed activation decisions z_s and realized demand q_s , the optimal assignment cost is captured by the recourse value:

$$\Phi_s(z_s, q_s) := \min_{x \in \{0,1\}} \sum_{(o,d) \in \Omega} q_{ods} \left(\sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \right) \quad (7)$$

subject to (2)–(3). Since this inner problem decomposes by OD pair, letting

$$m_{ods}(z_s) = \min_{\substack{(j,k) \in A_{od} \\ z_{js}=1, z_{ks}=1}} a_{odjks}$$

denote the cheapest admissible station pair where both pickup and dropoff are active in scenario s , the recourse value simplifies to

$$\Phi_s(z_s, q_s) = \sum_{(o,d) \in \Omega} q_{ods} m_{ods}(z_s). \quad (8)$$

The robust objective then minimizes the worst-case total cost across all scenarios:

$$\min_{y,z} \sum_{s \in S} \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s), \quad (9)$$

subject to (4)–(5). As (8) shows, Φ_s is linear in q_s for fixed z_s , so the inner maximisation over $q_s \in \mathcal{U}_s$ is a linear program in q_s and can be dualized to yield a tractable reformulation.

2.2.3 Derivation

Introducing epigraph variables η_s , the robust problem becomes

$$\min_{y,z,\eta} \sum_s \eta_s \quad (10)$$

subject to

$$\begin{aligned} \eta_s &\geq \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s) && \forall s, \\ z_{js} &\leq y_j && \forall j, s, \\ \sum_j y_j &= l, \quad \sum_j z_{js} = k && \forall s, \\ y_j, z_{js} &\in \{0, 1\}. \end{aligned}$$

Substituting (8) and dualizing the inner problem gives dual variables $\alpha_s \geq 0$, $\beta_{ods} \geq 0$, $\gamma_{ods} \geq 0$ such that

$$\eta_s \geq \bar{Q}_s \alpha_s + \sum_{(o,d)} \bar{q}_{ods} \beta_{ods} - \sum_{(o,d)} q_{ods} \gamma_{ods} \quad \forall s, \quad (11)$$

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq m_{ods}(z_s) \quad \forall (o, d), s. \quad (12)$$

Expanding $m_{ods}(z_s)$ with the assignment variables x and eliminating η_s yields the final MILP:

$$\min_{x,y,z,\alpha,\beta,\gamma} \sum_s \bar{Q}_s \alpha_s + \sum_s \sum_{(o,d)} \bar{q}_{ods} \beta_{ods} - \sum_s \sum_{(o,d)} q_{ods} \gamma_{ods} \quad (13)$$

subject to

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq \sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \quad \forall (o, d), s, \quad (14)$$

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s, \quad (15)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s, \quad (16)$$

$$z_{js} \leq y_j \quad \forall j, s, \quad (17)$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \quad \forall s, \quad (18)$$

with $\alpha_s, \beta_{ods}, \gamma_{ods} \geq 0$ and $x_{odjks}, y_j, z_{js} \in \{0, 1\}$.

3 Experiments

We use operational data from a microtransit system in Zhuzhou, China. The full network has 84 stations, but for scalability we reduce to the 40 most popular stations by OD request volume. Because these stations account for the majority of demand, the reduction in order coverage is minimal.

Table 1 describes the three variants. The sparse variant is constructed by randomly reassigning demand counts from the least popular OD pairs onto the most popular ones, concentrating all calibration demand onto the top 10% of OD pairs ($389 \rightarrow 130$). This mimics a data-poor setting where granular OD-level observations are unavailable and demand is only recorded at an aggregate level.

Table 1: Dataset variants for the April calibration and May backtest windows.

Metric	Full Zhuzhou	Sampled	Sparse
Stations	84	40	40
Directed segments	6,972	1,560	1,560
April orders	2,741	2,560	2,560
April ODs with orders	473	389	130
May orders	31,546	30,100	30,100
May ODs with orders	1,412	949	131

3.1 Experimental Design

We optimize station selection over data on April 1–3, 2025 and evaluate on May 1–31, 2025. We fix $l = 30$ selected stations, a 300-second walking limit, $\lambda = 10$, and vary $k \in \{10, 15, 20\}$. For the robust model we vary the calibration quantile $\tau \in \{0.90, 0.95, 0.99\}$.

3.2 Uncertainty-Set Calibration

Scenarios correspond to four time-of-day periods: morning (06–10h), afternoon (10–15h), evening (15–20h), and night (20–24h).

Lower bounds $\underline{q}_{ods}$: $\underline{q}_{ods} = 0$ for all OD pairs, since many pairs are absent on any given day.

Upper bounds $\bar{q}_{ods}$: Every OD pair in the full network receives a strictly positive upper bound $\bar{q}_{ods}$. For pairs observed at least once in the calibration window, $\bar{q}_{ods}$ is the τ -quantile of daily counts, treating days with no demand for that pair as zero. For pairs with no observed orders, $\bar{q}_{ods}$ is imputed via a rank-1 gravity model: $\bar{q}_{ods} = \lceil O_s(o) \cdot D_s(d) / Q_s \rceil$, where $O_s(o)$ and $D_s(d)$ are the total origin and destination flows from observed pairs and Q_s is the total observed demand in period s . For stations that appear in no orders at all, the gravity estimate falls back to a floor value equal to the minimum positive $\bar{q}$ observed across all periods.

Aggregate cap $\bar{Q}_s$: $\bar{Q}_s$ is the τ -quantile of total period demand across calibration days.

Table 2: Uncertainty-set diagnostics, 90th-percentile calibration.

Period	Sampled ODs	Sparse ODs	Sampled ratio	Sparse ratio
Morning (06–10h)	94	80	17.42	17.46
Afternoon (10–15h)	169	103	7.12	7.28
Evening (15–20h)	249	122	2.86	2.97
Night (20–24h)	120	87	13.16	13.31

Table 2 shows that both 40-station variants produce large box-to-cap ratios. This shows that the budget uncertainty set is a meaningful calibration of the uncertainty set. Suppose we were to not have the budget constraint, it would produce a solution that can be overly protective.

4 Results

Both experiments report results on the full-OD dataset (949 OD pairs), covering both the April calibration period (2,560 orders) and the May test period (30,100 orders). For the sparse experiment, this means evaluating on OD pairs not seen during calibration: models were trained on 130 OD pairs while evaluation spans all 949.

4.1 Sampled-Network Results

With dense training OD coverage, all models achieve zero uncovered orders (Table 3). The nominal model has the lowest mean cost for every k ; robust variants trade a small mean cost increase for lower variance.

Table 3: Sampled-network results, full-OD May orders. Bold = best in column within each k -block.

k	τ	Model	April mean cost	May mean cost	April std.	May std.	Uncovered
10	–	Nominal	4266.09	4333.42	1129.72	1116.39	0
10	0.90	Robust	4287.81	4359.76	1111.08	1102.67	0
10	0.95	Robust	4302.24	4392.66	1108.89	1090.17	0
10	0.99	Robust	4293.94	4369.71	1109.24	1098.48	0
15	–	Nominal	3977.09	4049.94	1145.30	1119.97	0
15	0.90	Robust	4018.31	4071.27	1106.48	1088.71	0
15	0.95	Robust	4023.38	4080.47	1105.51	1085.45	0
15	0.99	Robust	4030.38	4067.03	1096.25	1066.90	0
20	–	Nominal	3924.95	3984.77	1168.67	1142.49	0
20	0.90	Robust	3967.29	4003.44	1116.51	1082.44	0
20	0.95	Robust	3948.96	3978.06	1118.75	1090.00	0
20	0.99	Robust	3958.28	4005.45	1118.56	1091.22	0

4.2 Sparse-Network Results

The nominal model fits the sparse training data best but leaves 227 orders uncovered at $k = 10$ and 14 at $k = 15$. Robust models achieve complete coverage (Table 4). At $k = 20$ all models cover every order and costs converge.

All three models select the same 30 stations (see Figure 3 in the appendix), so the divergence is entirely in which k stations are activated per period. Figure 1 shows the morning period at $k = 10$: the nominal model places all 10 activations in the high-demand core near the centre of the network, while the robust ($q = 0.95$) model distributes activations more broadly, including peripheral stations that are rarely observed in the sparse calibration data. This broader coverage is what eliminates the 227 uncovered orders.

However, for orders that *are* covered, the cost distributions are nearly identical. Figure 2 shows that CDFs for nominal and robust largely overlap, with robust variants shifted only slightly to the right in the upper tail. This reveals a gap between the expected and actual benefit of robustness: the model was designed to hedge against worst-case costs, but in practice the recourse function is simple enough that once a station is within reach, the assignment cost is similar regardless of which model selected the stations. The main benefit in the sparse setting is therefore service availability,

Table 4: Sparse-network results, full-OD May orders. Bold = best in column within each k -block.

k	q	Model	April mean cost	May mean cost	April std.	May std.	Uncovered
10	–	Nominal	4446.97	4305.57	851.46	1092.41	227
10	0.90	Robust	4505.20	4363.35	845.97	1100.00	0
10	0.95	Robust	4521.04	4392.62	836.33	1094.39	0
10	0.99	Robust	4507.51	4368.66	840.04	1089.52	0
15	–	Nominal	4182.48	4031.55	881.21	1082.65	14
15	0.90	Robust	4234.00	4080.66	853.99	1059.06	0
15	0.95	Robust	4235.88	4089.67	848.30	1058.01	0
15	0.99	Robust	4236.58	4073.98	855.82	1067.54	0
20	–	Nominal	4136.13	3972.53	901.73	1094.37	0
20	0.90	Robust	4149.90	3985.01	886.97	1082.80	0
20	0.95	Robust	4155.78	3998.38	883.00	1077.17	0
20	0.99	Robust	4148.22	3984.41	887.77	1079.39	0

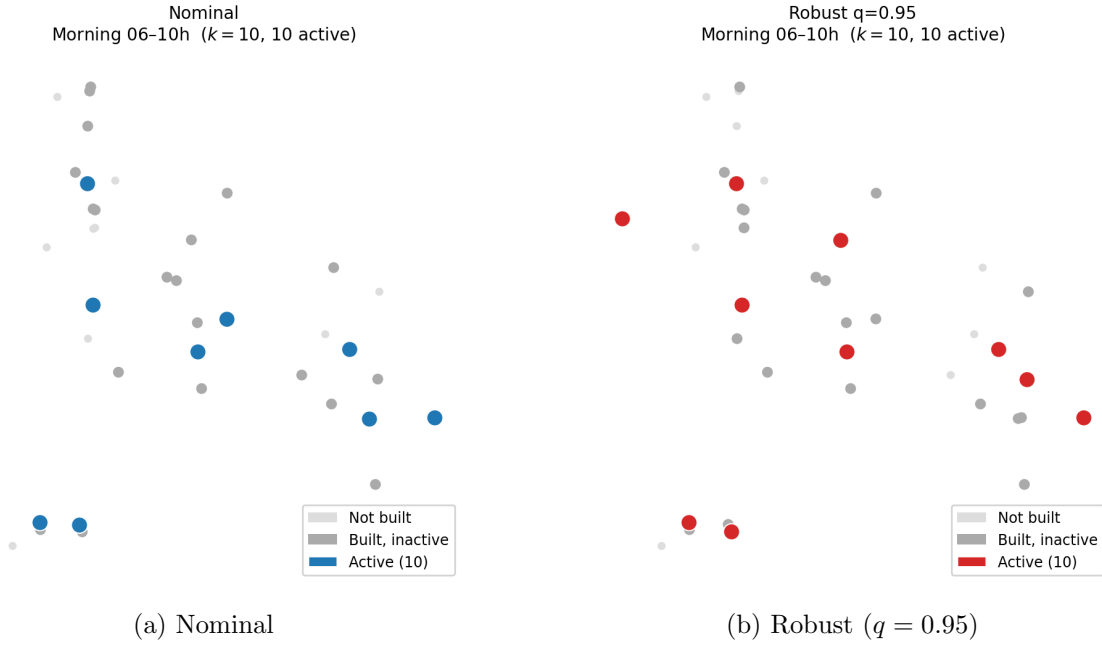


Figure 1: Morning-period activations for $k = 10$. Filled = active; hollow = selected but inactive. Nominal clusters all activations in the high-demand core; robust spreads them across the network.

specifically ensuring that peripheral demand is not left uncovered, rather than cost reduction for served orders.

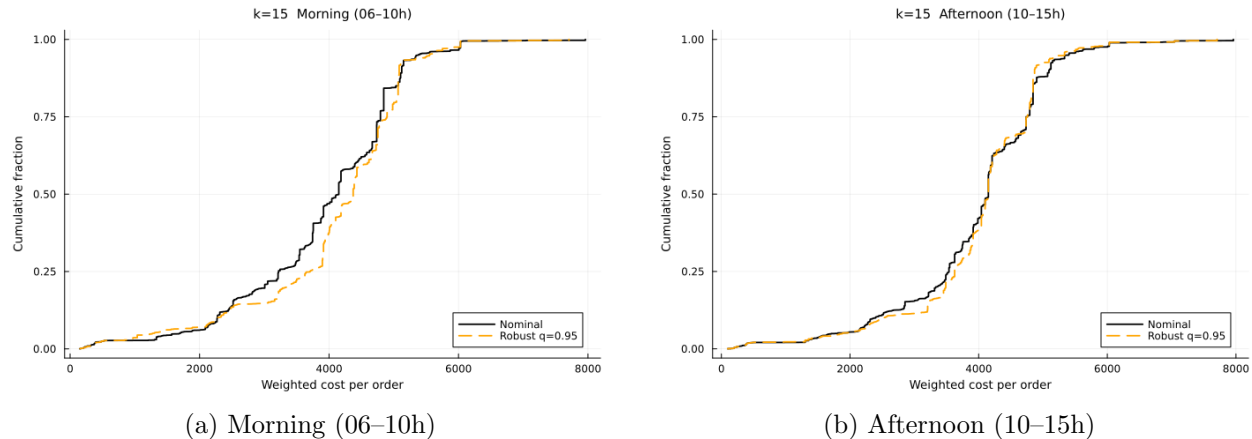


Figure 2: Cost CDFs for $k = 15$, sparse experiment, May full-OD test set. Nominal and robust distributions largely coincide for covered orders.

5 Conclusion

A budgeted uncertainty set is a practical way to model uncertain OD demand without extreme conservatism: per-OD bounds keep each pair within its historical range while the cap $\bar{Q}_s$ prevents simultaneous surges. When data is dense, robustness primarily reduces cost variance. When data is sparse, the nominal plan concentrates activations in the observed high-demand core and leaves peripheral orders uncovered, while the robust plan achieves full coverage at a modest cost premium. Coverage can alternatively be enforced via explicit constraints, which may be more efficient when variance reduction is not a goal.

Future work could incorporate richer recourse functions reflecting true microtransit operations (vehicle capacity, rebalancing, multi-period dynamics), and identify which demand regimes, such as low-density or spatially shifting demand, most benefit from a robust plan.

A Additional Figures

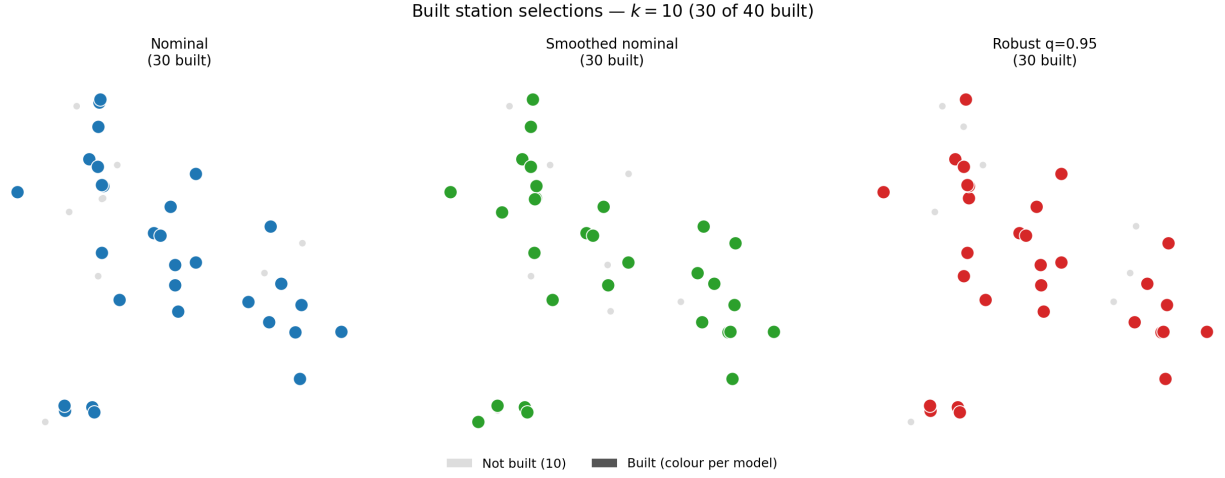


Figure 3: Selected stations for $k = 10$. Coloured dots = selected (30 of 40); gray = not selected. All three models select the same set of 30 stations.

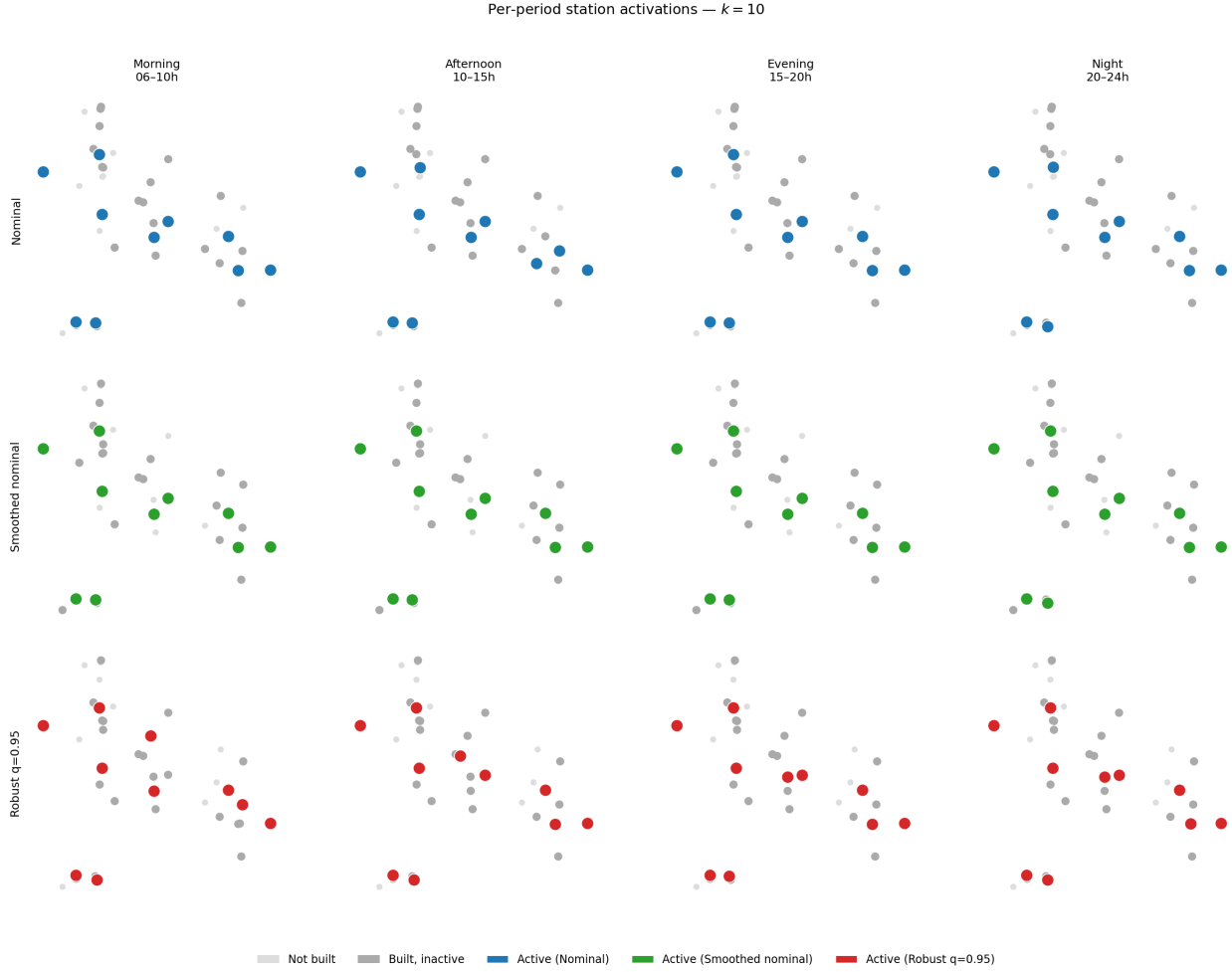
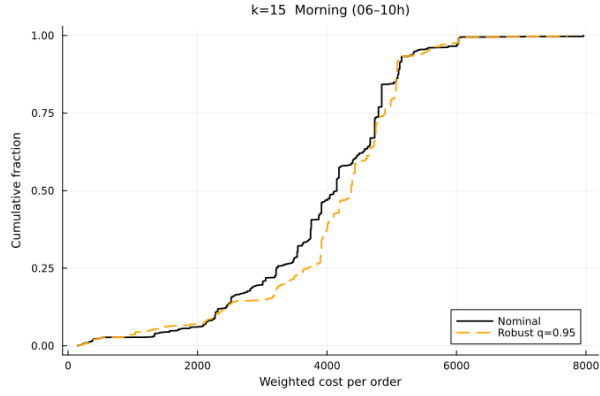
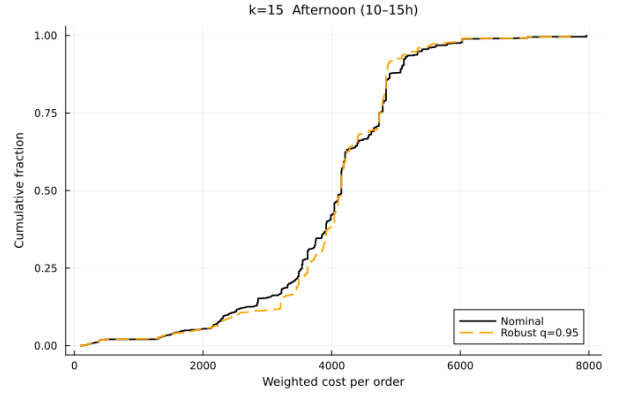


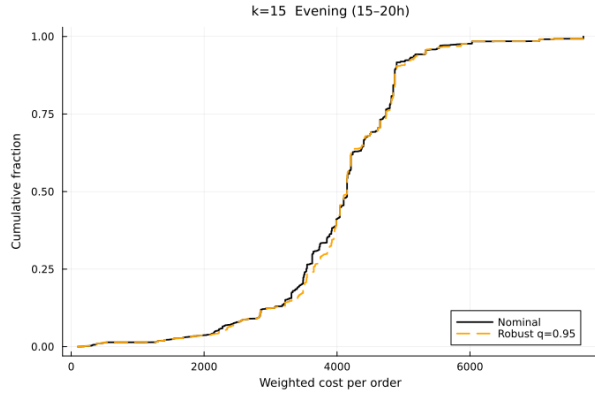
Figure 4: Per-period activations for $k = 10$ across all four time-of-day scenarios. Filled = active; hollow = selected but inactive; gray = not selected.



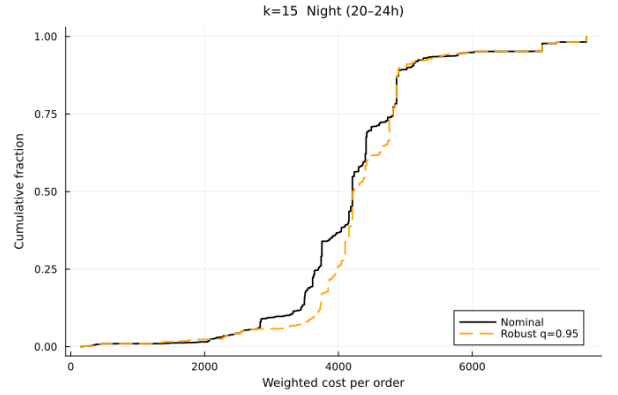
(a) Morning (06-10h)



(b) Afternoon (10-15h)



(c) Evening (15-20h)



(d) Night (20-24h)

Figure 5: Cost CDFs for $k = 15$ across all four time-of-day scenarios, sparse experiment, May full-OD test set.